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| **Begonnen am** | Thursday, 16. December 2021, 10:30 |
| **Status** | Beendet |
| **Beendet am** | Thursday, 16. December 2021, 10:50 |
| **Verbrauchte Zeit** | 20 Minuten 5 Sekunden |

Frage **1**
Vollständig
Erreichbare Punkte: 5,00

The following are examples where machine learning is used for visualization (ML4Vis)

- [ ] a. a visualization showing the latent space of a deep neural network
- [x] b. a machine learning model used for detecting recurring patterns in time-series data
- [ ] c. a visualization showing potentially interesting patterns identified by a pre-trained model
- [x] d. a machine learning model used for suggesting suitable plot types for a given dataset

Frage **2**
Vollständig
Erreichbare Punkte: 5,00

The following are valid statements for visualizing time-series (time-dependant) data

- [x] a. animation can make it hard to compare data from different time points
- [x] b. the progression of the data over time can be shown using animation
- [x] c. the data at different time points can be shown next to each other
- [x] d. the time aspect can be encoded using color or other visual channels

Frage **3**
Vollständig
Erreichbare Punkte: 5,00

The following are valid statements in the context of deep learning visualization

- [ ] a. heatmaps are commonly used for showing spatial attribution
- [ ] b. stream graphs are commonly used for showing the architecture of a network
- [ ] c. heatmaps are commonly used for showing instance-specific features
- [x] d. heatmaps are commonly used for showing confusion between classes

Frage **4**
Vollständig
Erreichbare Punkte: 5,00

The following are disadvantages of dimensionality reduction techniques

- [x] a. visual patterns in the lower dimensional embedding space can be misleading
- [ ] b. outliers in the data cannot be taken into account
- [ ] c. categorical attributes/dimensions can be taken into account
- [x] d. it can be difficult to preserve the semantics of single attributes/dimensions

Frage **5**
Vollständig
Erreichbare Punkte: 5,00

The following statements are valid in the context of data-driven storytelling

- [ ] a. a purely reader-driven story orders the parts of the story in a specific way
- [x] b. a purely author-driven story does not allow the reader to interact with the information
- [ ] c. a purely reader-driven story does not allow the reader to interact with the information
- [ ] d. a drill-down story starts author-driven and then opens up for exploration

Frage **6**
Vollständig
Erreichbare Punkte: 5,00

The following statements are valid for the data-ink ratio by Edward Tufte

- [ ] a. should be 0 for producing effective visualizations
- [x] b. chart junk increases the ratio
- [x] c. should be understood as a general guideline rather than a strict optimization criterion
- [x] d. should be maximized for producing effective visualizations

Frage **7**
Vollständig
Erreichbare Punkte: 5,00

3D is known to be problematic for visualizing multi-dimensional numerical data because of

- [ ] a. animated transitions
- [x] b. the necessity to rotate the visualization for inspecting hidden data
- [ ] c. inattentional blindness
- [x] d. perspective distortion effects

Frage **8**
Vollständig
Erreichbare Punkte: 5,00

A storytelling article that has multiple predefined substories and where the reader can decide which one to follow is known as a

- [ ] a. Whisky glass structure
- [ ] b. Martini glass structure
- [ ] c. Scrollytelling interface
- [x] d. Drill-down story

Frage **9**
Vollständig
Erreichbare Punkte: 5,00

Which of the following statements is/are valid in the context of grouping principles?

- [x] a. bump charts rely on the connection grouping principle
- [ ] b. the continuation grouping principle overrules similarity and proximity
- [x] c. clusters of items in scatterplots rely on the proximity grouping principle
- [ ] d. scatterplots rely on the continuation grouping principle

**Frage 10**
Vollständig
Erreichbare Punkte: 5,00

The following are valid approaches for handling large datasets

- [ ] a. aggregate the data before visualizing it
- [x] b. combine visualization with algorithmic/automated techniques
- [x] c. allow users to interact with the data
- [x] d. sample the data before visualizing it

**Frage 11**
Vollständig
Erreichbare Punkte: 5,00

The following are valid statements for the InstanceFlow visualization technique

- [ ] a. shows the evolution of instance class confusion using a Sankey diagram
- [x] b. is able to show the classification of single instances over multiple epochs
- [x] c. is able to show which parts of images contribute to the classification result
- [ ] d. is able to rank single instances by various scores/measures

**Frage 12**
Vollständig
Erreichbare Punkte: 5,00

The following are properties of the t-SNE dimensionality reduction

- [ ] a. random noise in the data looks random in the embedding space
- [ ] b. cluster sizes in the embedding space are meaningless
- [x] c. distances between clusters in the embedding space can be meaningless
- [ ] d. shapes in the data are preserved in the embedding space

**Frage 13**
Vollständig
Erreichbare Punkte: 5,00

For Projective Latent Interventions, a parametric dimension reduction technique is necessary because

- [ ] a. it needs to be applied to new activation vectors that were not part of the original embedding
- [ ] b. the dimensionality of the activations is too high for standard techniques
- [ ] c. it allows backpropagation of an additional error term
- [x] d. standard techniques are computationally too inefficient

**Frage 14**
Vollständig
Erreichbare Punkte: 5,00

The following are valid statements for the Time Curves technique

- [x] a. the temporal ordering of the data is preserved
- [ ] b. the layout is independent of the projection method used
- [ ] c. can be applied to visualize changes in video data
- [ ] d. spatial proximity in the visualization indicates similarity between time points

Frage **15**
Vollständig
Erreichbare Punkte: 5,00

The following visualization techniques can typically be used for representing the functioning of sorting algorithms

- [ ] a. the Matchmaker technique
- [ ] b. attribution maps
- [ ] c. heatmaps
- [x] d. bump charts

Frage **16**
Vollständig
Erreichbare Punkte: 5,00

The following statements are valid for particle-based visualizations

- [ ] a. are based on sedimentation effects that can be observed in nature
- [x] b. are regularly used for visualizing Bayes' theorem
- [ ] c. are regularly used for showing outliers in time-series data
- [x] d. can be used to show progressive changes in streaming data

Frage **17**
Vollständig
Erreichbare Punkte: 5,00

Projective Latent Interventions allow users to

- [ ] a. re-label individual instances on the fly
- [x] b. monitor the accuracy of a classification model
- [ ] c. monitor the accuracy of a regression model
- [ ] d. shape the latent space of a classification model

Frage **18**
Vollständig
Erreichbare Punkte: 5,00

The following statements are valid for Anscombe's quartet

- [ ] a. consists of five datasets with two dimensions each
- [x] b. is used as a motivating example for the value of visualization
- [x] c. the datasets share important statistical properties
- [x] d. the datasets look very different when being plotted

Frage **19**
Vollständig
Erreichbare Punkte: 5,00

A k-means clustering result shown in a scatterplot can be visualized by

- [ ] a. modifying the position of items according to their cluster assignment
- [x] b. assigning different shapes to items by cluster
- [ ] c. connecting all cluster centroids with each other
- [x] d. connecting items to the cluster centroids they belong to

Frage **20**

Vollständig

Erreichbare Punkte: 5,00

The following are valid statements for the ConfusionFlow visualization technique

- [ ] a. does not provide any global performance measures
- [ ] b. shows the evolution of class confusion using a stream graph
- [x] c. operates on the class level
- [ ] d. shows the evolution of class confusion using a Sankey diagram

◀ 2021 XAI Lab Assignment 2 Introduction - Recording

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2021 VA Lab Plotting in Python - Static Plotting ▶